

Fundamentals of Solid State Physics

Introduction

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Goal of This Course

- Study the foundations of our world: **solid materials**
 - electronic, optical, magnetic, thermal, mechanical, ...
- Based on **theories of physics**
 - classical mechanics
 - electrodynamics
 - quantum mechanics
 - statistical mechanics
 - ...
- **Emerging applications**: semiconductors, lasers, ...

Goal of This Course

- Main Focus: **Electrons**

Department of **Electronic** Engineering
电子工程系

- Other topics: Photons, Phonons, Magnetics, ...

What is Unique?

- Programming (程序)
- Circuits (电路)
- Data and Algorithm (数据与算法)
- Signals and Systems (信号与系统)
- Probability (概率)
- Digital Logics (数字逻辑)
- Communications (通信)
- Media and Cognition (媒体与认知)
- Electromagnetism (电磁场)
- **Solid State Physics (固体物理)**

**10 core courses
in EE department**

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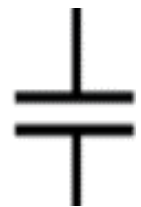
$$v = iR$$



$$v = L \frac{di}{dt}$$



$$i = C \frac{dv}{dt}$$



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$$\nabla \cdot \mathbf{D} = \rho_V$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$

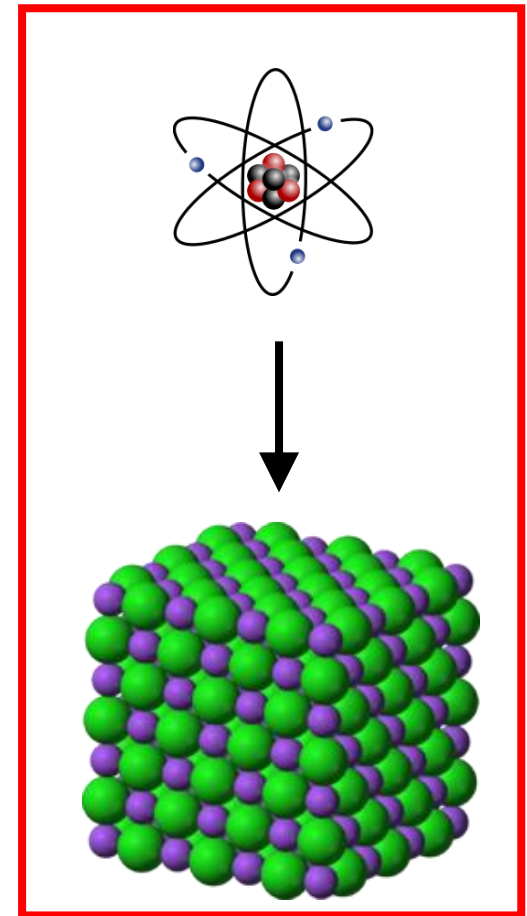
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maths,
design,
engineering,
application,
model,
perfect,
...

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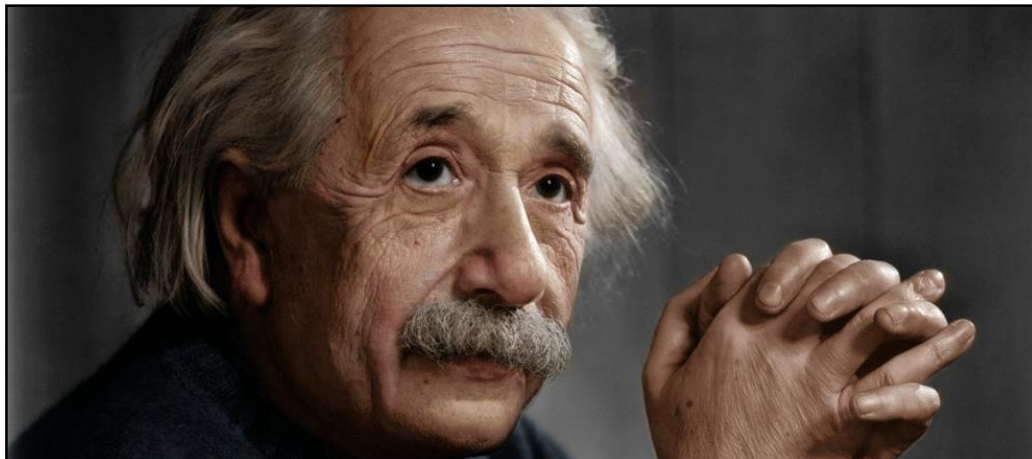
**observation,
discovery,
understanding,
approximation,
assumption,
modified theory,
imperfection,
...**

Course Philosophy

"The eternal mystery of the world is its comprehensibility. ... The fact that it is comprehensible is a miracle."

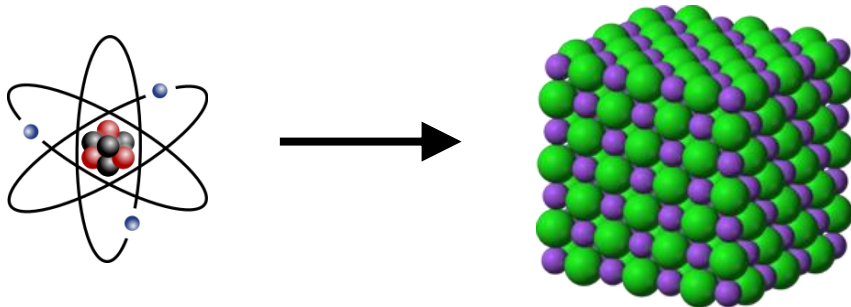
这个世界最不可理解之处在于它是可以被理解的

---- Albert Einstein



Solid is very Complex

- If we understand all the basic physics of the fundamental particles (electrons, photons, atoms, ...), can we understand everything about solids?



No!

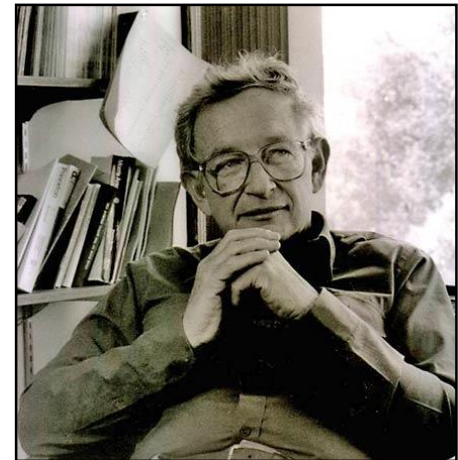
ARTICLES

More Is Different

By P. W. Anderson

+ See all authors and affiliations

Science 04 Aug 1972:
Vol. 177, Issue 4047, pp. 393-396
DOI: 10.1126/science.177.4047.393

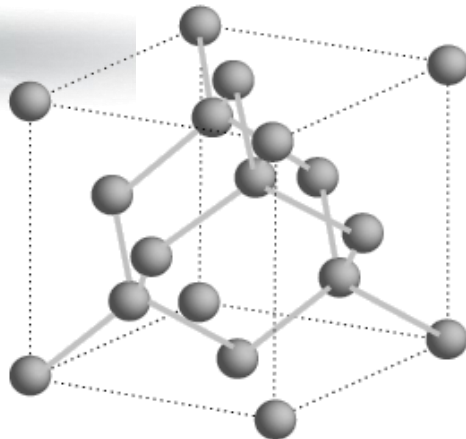


Philip W. Anderson
1923–2020

Example: Carbon

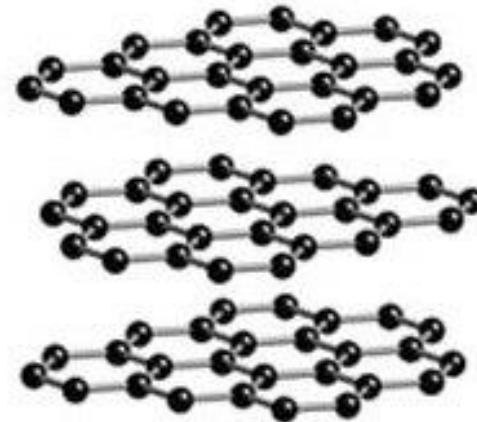
■ Diamond 金刚石

- ❑ hard
- ❑ insulating
- ❑ transparent



■ Graphite 石墨

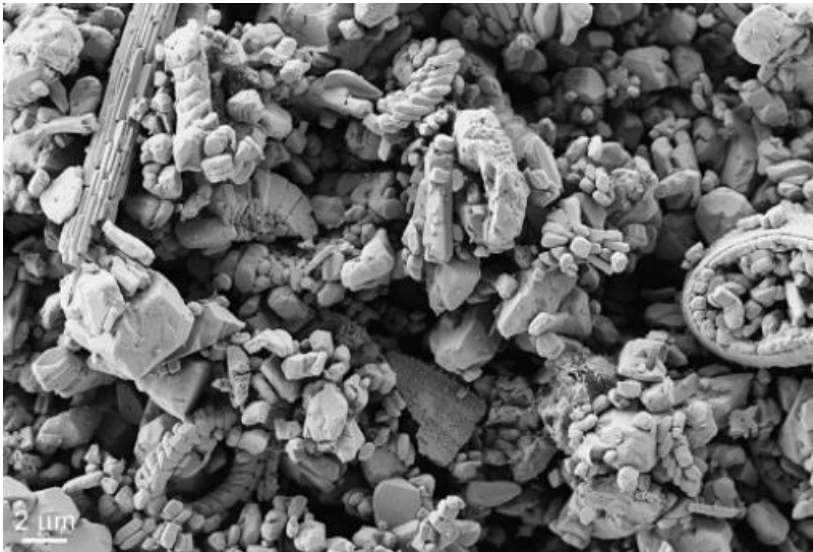
- ❑ soft
- ❑ conductive
- ❑ black



Example: CaCO_3

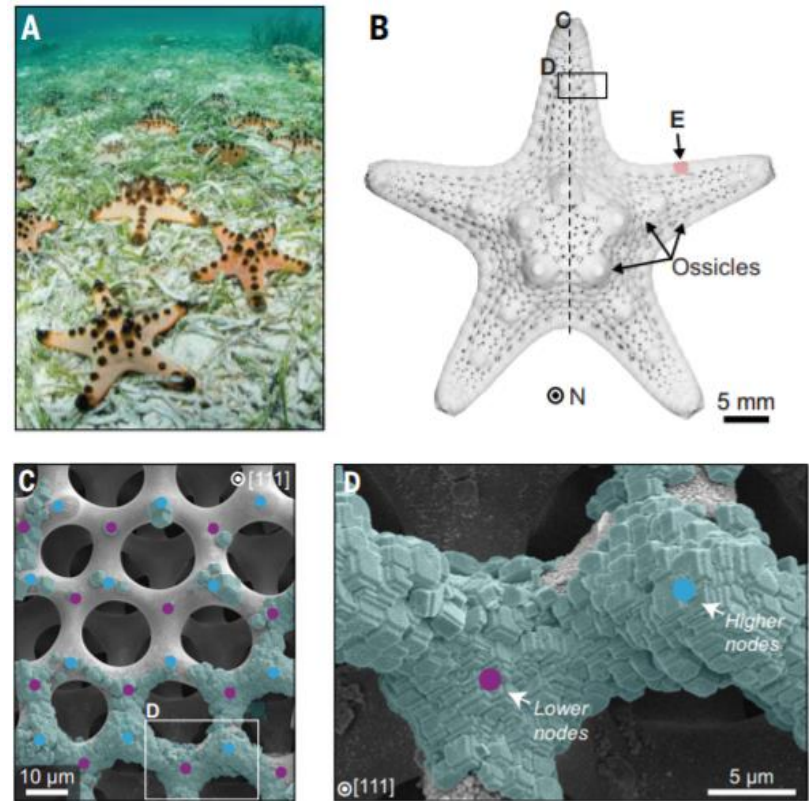
■ Chalk 粉笔

□ brittle



■ Starfish 海星

□ robust



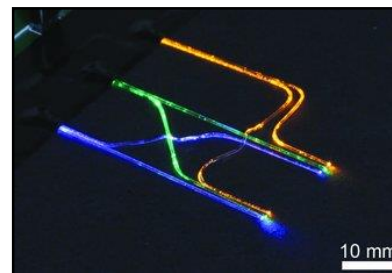
Optical and Electronic Devices



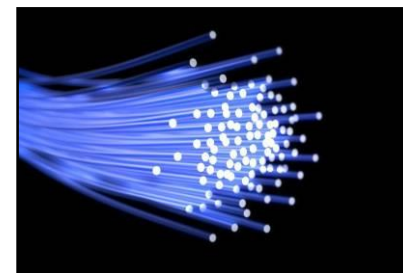
LEDs



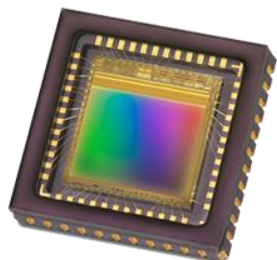
lasers



waveguides



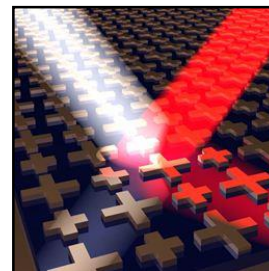
fibers



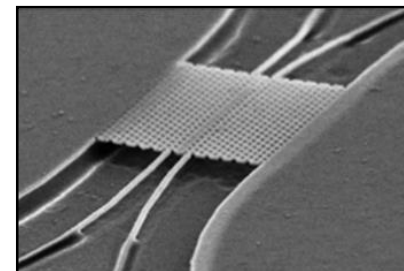
detectors



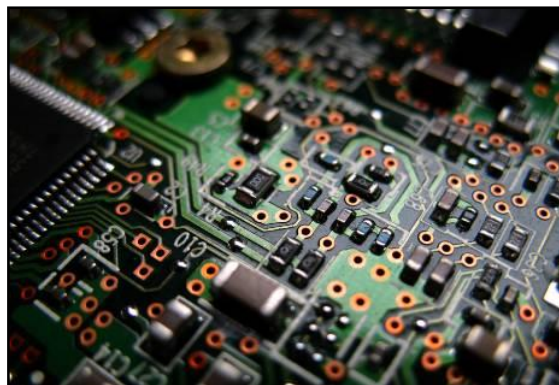
solar cells



metamaterials



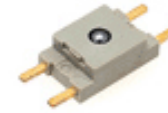
photonic crystals



integrated circuits

Airflow
SensorsCurrent
SensorsFiber Optics and
Liquid Level Sensors

Flexible Heaters



Force Sensors

Humidity
Sensors

Infrared Sensors

Magnetic
SensorsProximity
SensorsRotary Position
SensorsSpeed
Sensors

Nobel Prizes in Solid-State Physics

- Before 1950s
 - ▣ atoms, materials, radiation, quantum mechanics, ...
- 1956 Semiconductor transistors
- 1991 Liquid crystals
- 2000 Integrated circuits
- 2009 CCD imaging sensors
- 2009 Optical fibers
- 2014 GaN based blue LEDs

'Disruptive' Technologies 颠覆性技术

- Before 1950s
 - ▣ atoms, materials, radiation, quantum mechanics, ...
- 1956 Semiconductor transistors ← vacuum tubes
- 1991 Liquid crystals ← CRT displays
- 2000 Integrated circuits ← Computation
- 2009 CCD imaging sensors ← film cameras
- 2009 Optical fibers ← copper cables
- 2014 GaN based blue LEDs ← incandescent light bulbs

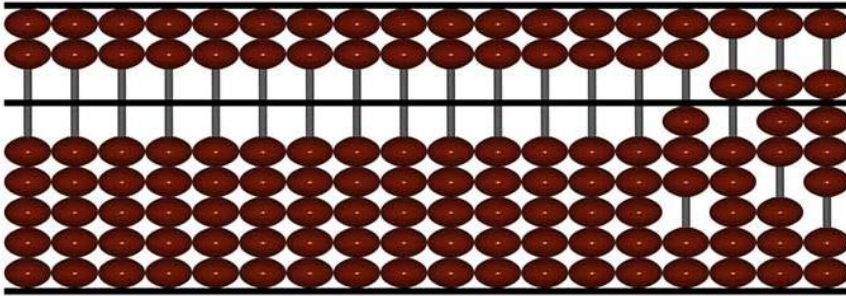
Devices in a Smartphone



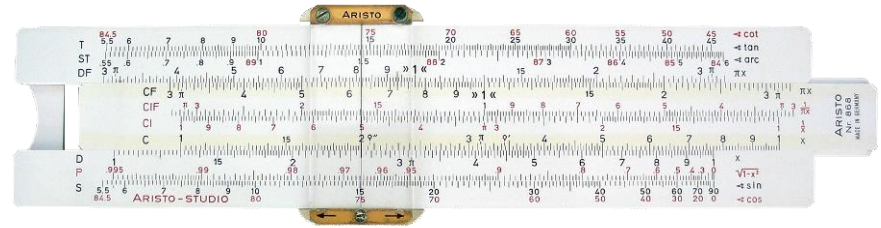
Nobel Prize

■ Transistors	1956
■ Liquid crystals	1991
■ Integrated circuits	2000
■ Semiconductor heterostructures	2000
■ CCD cameras	2009
■ GaN blue LEDs	2014
■ Li ion batteries	2019 (Chemistry)

Computation

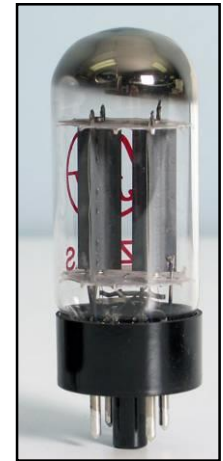
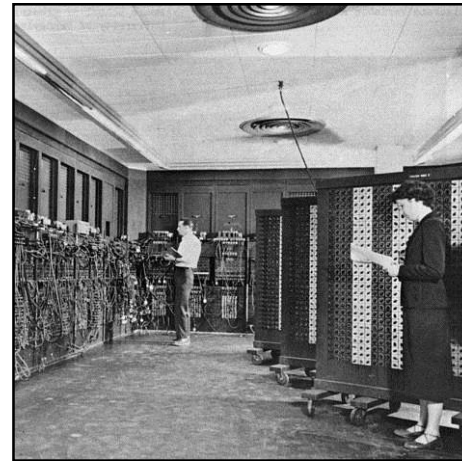


abacus 算盘



slide rule 计算尺

- Ancient computers
- First 'electronic' computer
 - ❑ ENIAC, 1943
 - ❑ 30 tons, 200 kW
 - ❑ 18000 vacuum tubes
 - ❑ 5000 times/sec
 - ❑ cost \$480,000



vacuum tube
电子管/真空管 28

Transistors

Germanium
Bipolar Transistor



The first point contact transistor
William Shockley, John Bardeen, and Walter Brattain
Bell Laboratories, Murray Hill, New Jersey (1947)

semiconductors

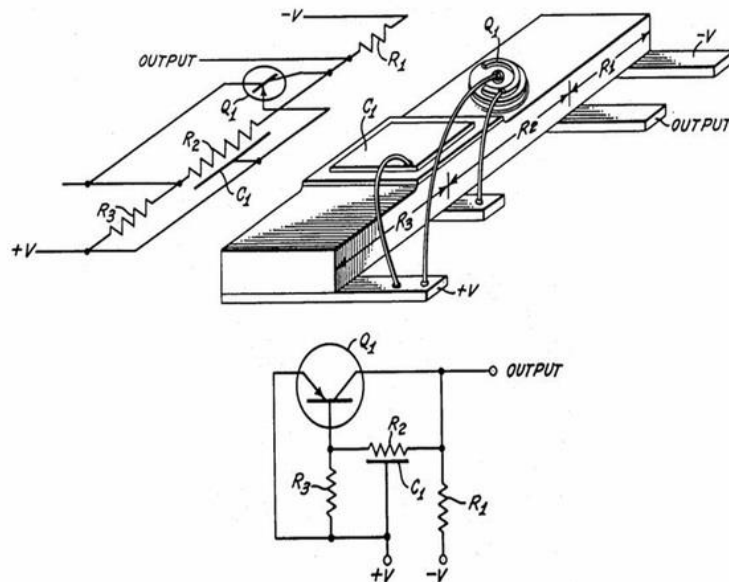


1956 Nobel Prize in Physics

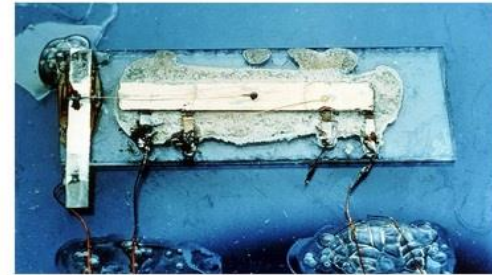
First Integrated Circuits

The First (2D) Integrated Circuit Jack Kilby, Texas Instruments, 1958

- Transistor, Resistors and Capacitors on the same piece of semiconductor
- Interconnects between components not integrated
→ Low connectivity between components



Germanium

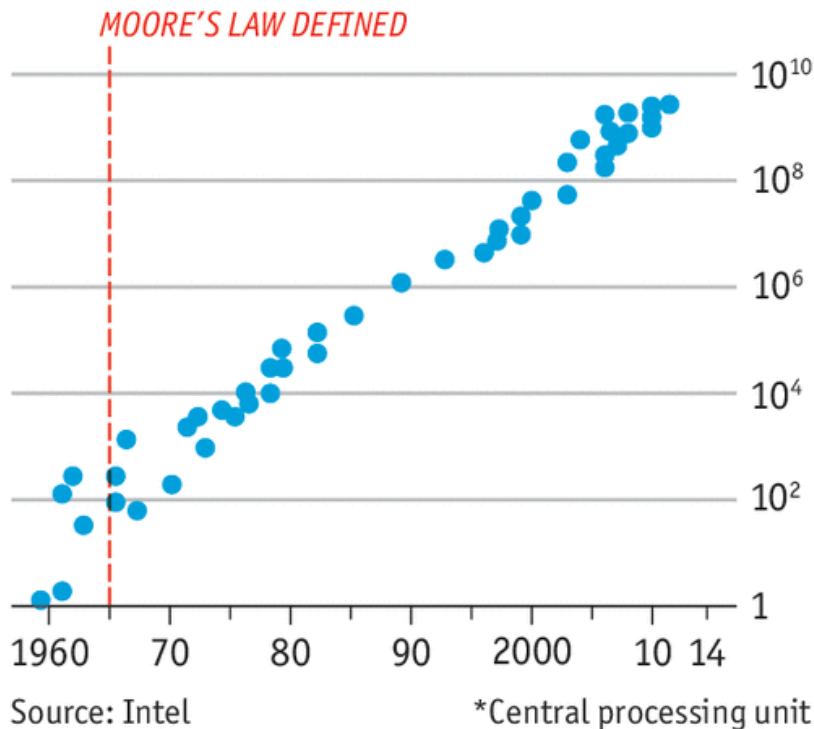


J. Kilby

2000 Nobel Prize in Physics

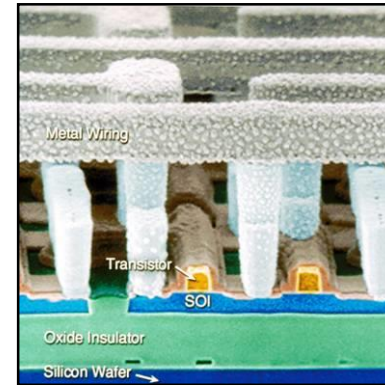
Modern Integrated Circuits

■ Moore's law, Fairchild, 1965

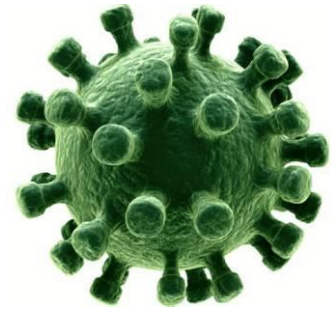


Economist.com

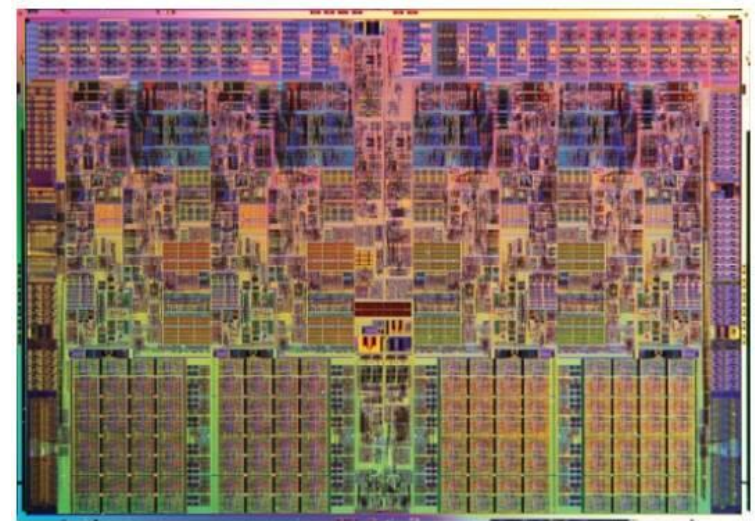
***Modern Electronics is a
real Nanotechnology***



< 100 nm



coronavirus



Intel i7 CPU, ~ 10^9 transistors

Lighting



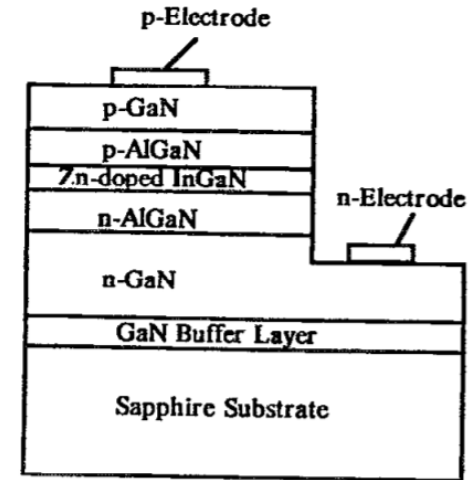
Incandescent bulb



Fluorescent lamp



GaN blue LEDs



S. Nakamura, et al., *Appl. Phys. Lett.* **64**, 1687 (1994)



I. Akasaki



H. Amano

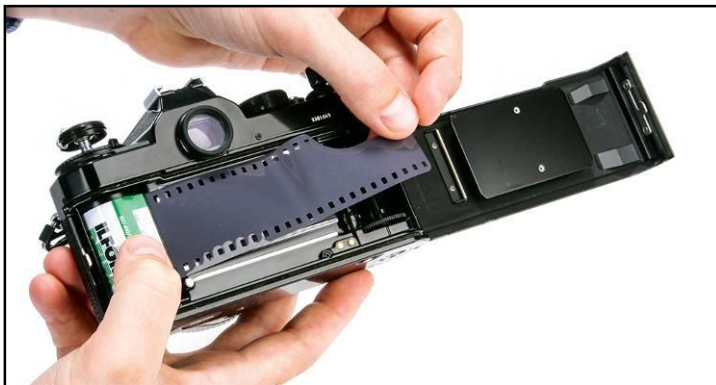


S. Nakamura

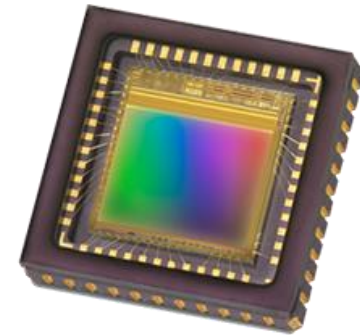
2014 Nobel Prize in Physics 32

Imaging

CCD and CMOS cameras



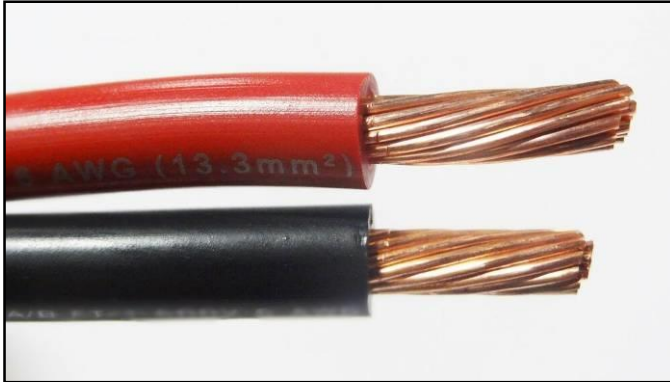
films



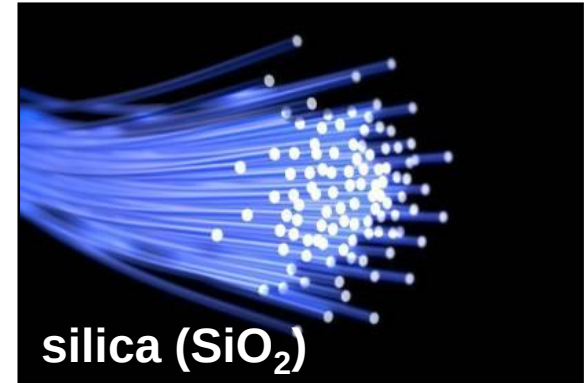
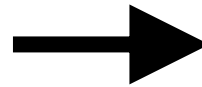
W. Boyle and G. Smith

2009 Nobel Prize in Physics 33

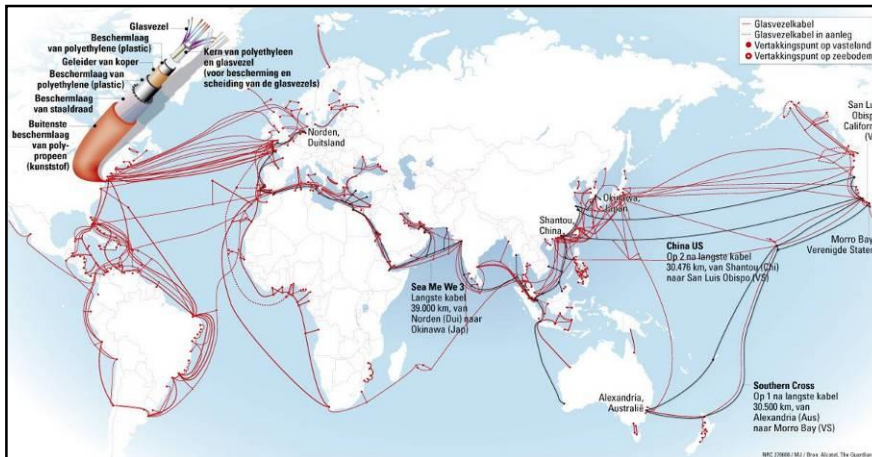
Communication



copper cables



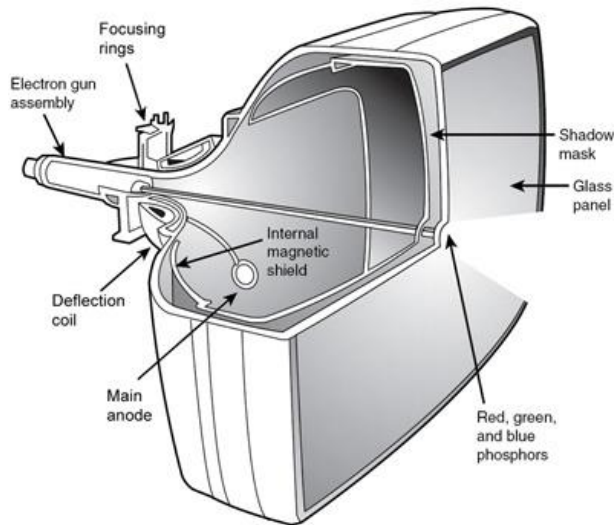
optical fibers



K. Kao (高錕)

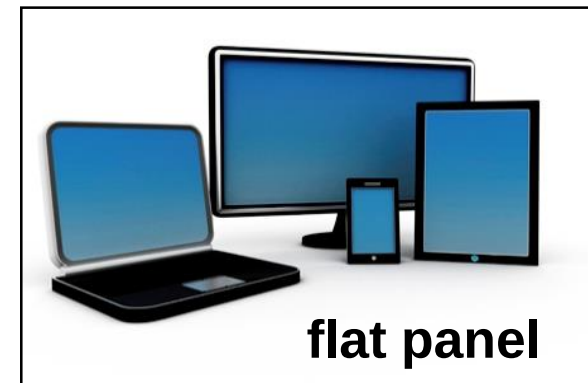
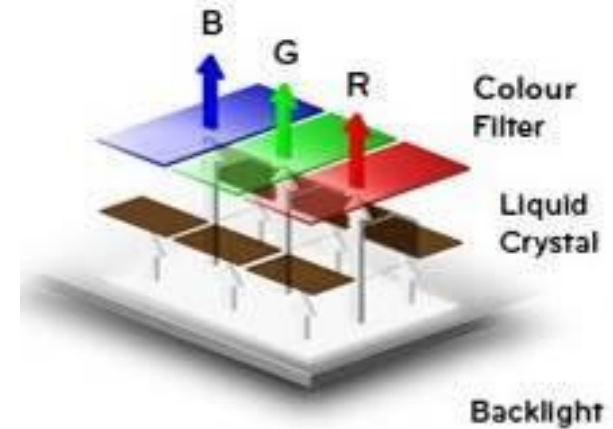
2009 Nobel Prize in Physics

Displays



Cathode Ray Tube (CRT)

liquid crystal display (LCD)



flat panel

de Gennes
1991 Nobel Prize in Physics

Syllabus

- **Introduction (Week 1)**
- **Materials and Crystal Structures (Week 2–3)**
- **Electronic Properties (Week 4–12)**
- **Thermal Properties (Week 13)**
- **Optical Properties (Week 14)**
- **Magnetic Properties (Week 15)**

Syllabus

- **Introduction (Week 1)**
 - ▣ Overview, history, applications
 - ▣ Preliminary knowledge
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- **Materials and Crystal Structures (Week 2–3)**
 - ▣ Bravais lattices, Crystal structures, Defects
 - ▣ Reciprocal space, Brillouin zones
 - ▣ Materials Characterization: Wave diffraction, Bragg's law
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 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential, Bloch's Theorem
 - The nearly free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass, holes
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
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 - ▣ Crystal vibration, phonon band
 - ▣ Thermal conductivity and capacity
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- **Optical Properties (Week 14)**
 - ▣ **Origin of Dielectric constant (ϵ) and Refractive index (n)**
 - ▣ **Optical absorption, reflection, refraction, emission**
- Magnetic Properties (Week 15)

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 - **Diamagnetism, Paramagnetism, Ferromagnetism**
 - **Superconductivity**

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